

Strata — Developer Guide

For engineers implementing tickets

strata-kb v1.0.1 · 2026-09-19

1. Who this guide is for

You write product code. Tickets arrive already grounded in the organisation's knowledge base, with citations pinned to a specific version of a specific document, and you implement them.

Strata does two things for you.

It tells you what the source actually said. Not "check the handbook" — the exact section, at the exact revision the analyst was reading, with a warning if it has changed since.

It publishes what your codebase is, so everyone else — analysts drawing architecture diagrams, other teams, future you — can search your services, dependencies, endpoints and schema without reading your repository.

1.1 What a dev repo does and does not do

Does	Does not
Read the hub's published knowledge	Ingest documents from outside the repo
Publish knowledge about its own code	Publish domain knowledge
Run a five-command implementation workflow	Merge or ship anything by itself

The boundary is deliberate. A `child` repository brings in outside documents; a `dev` repository ingests nothing and authors only knowledge about itself. That is what makes it safe for your CI to publish continuously.

1.2 The two documents your repo publishes

Document	Origin	Regenerated	Needs review
<code><repo_id>-code</code>	<code>kb code-ingest</code> — deterministic, no model	every push to the default branch	no
<code><repo_id>-svc</code>	drafted by a model from that evidence, corrected by you	never	always

`-code` is machine fact: service names, technologies, dependencies, commands, endpoints, schema, folder structure.

`-svc` is human claim: what each service is actually *responsible for*.

They are joined by section id — `svc.<name>` exists in both — so an analyst drawing a C4 container gets the alias, label and technology from `-code` and the description from `-svc`.

2. Setup, once

```
pip install strata-kb
kb init --kind dev
```

2.1 Configure

```
# .kb/config.yaml
kind: dev
repo_id: payments
hub: https://github.com/acme/kb-hub.git
intake: https://kb-intake.acme.com
```

- `hub` is the only read source for `kb query`, `kb get`, `kb resolve` and MCP.
- `intake` is how CI publishes. Ask the hub maintainer to add this repository under `repos:` in the hub's `federation/registry.yaml` — publishing is refused until they do.

2.2 Connect your assistant

`.mcp.json` (Claude Code) and `.cursor/mcp.json` (Cursor) were scaffolded already wired to the hub's MCP tools, but through placeholders — `${STRATA_KB_HUB_URL}` and `${STRATA_KB_HTTP_TOKEN}` — that nothing has set yet.

Run `kb mcp-setup` (in your assistant: `/kb-mcp-setup`). It asks for the hub's HTTP base URL and token, writes both into `.env`, makes sure git ignores that file, and then verifies them against the hub, so a wrong URL and a rejected token come back as different errors:

Variable	Example
<code>STRATA_KB_HUB_URL</code>	<code>http://kb-hub.example.com:8321</code>
<code>STRATA_KB_HTTP_TOKEN</code>	the hub's bearer token

A bare re-run verifies again without retyping the token. Prefer the hidden prompt or the `STRATA_KB_HTTP_TOKEN` environment variable over `--token` — that flag leaves the token in your shell history.

`.mcp.json` and `.cursor/mcp.json` read those two variables from your environment, not from `.env` directly, so load `.env` into your shell (`set -a; source .env; set +a`, or use `direnv`) and restart your assistant — MCP reads the environment only at startup.

2.3 Turn on the PR gate

`kb init` writes `.github/workflows/kb-pr-lint.yml` but cannot enable branch protection for you. Add `pr-lint` to the branch's required checks.

Decide this knowingly. Unlike the BA repo's gate, this one never self-skips. Once required, it blocks **every** pull request lacking the eight required sections — including bot pull requests such as a dependency bump or a revert.

2.4 Ask about the hub's auto-merge policy

`-code` pull requests on the hub may be auto-merged; `-svc` pull requests must never be. Ask your hub maintainer how it is configured, and make sure of one thing:

`kb ci-publish` opens **one** hub pull request per push, not one per document. The intake keys the publish branch by repository id and reuses it while a PR is pending — so a push carrying a regenerated `-code` alongside an already-merged `-svc` amend lands **both** in that same PR.

The auto-merge rule must therefore be **path-scoped to** `.kb/<repo_id>-code/**`, never a whole-PR or whole-repo rule. A rule that fires on "this is a dev repo's `-code` publish" silently defeats the `-svc` review gate the two-document split exists to protect.

3. Onboarding an existing project

Your repository already has services doing real work. Adopting Strata should not mean documenting the system from a blank page, so there is a one-time seed.

```
/dev-code-seed
```

Seven steps:

#	Step	Who
1	Preflight — confirms <code>hub:</code> , <code>repo_id:</code> and <code>intake:</code> are set and the repo is allowlisted on the hub; warns on a dirty tree	agent
2	Extract — <code>kb code-ingest --scaffold-svc</code> regenerates <code>-code</code> and creates a <code>pending svc.<name></code> scaffold per detected service, backed by deterministic code evidence	agent
3	Draft — <code>kb summarize <repo_id>-svc</code> writes a responsibility paragraph for each from that evidence	agent
4	Review — walk every drafted section against its evidence and correct it	you
5	Approve — <code>kb approve <repo_id>-svc</code> flips corrected sections to <code>reviewed</code>	you
6	Flows (optional) — add <code>flow.<name></code> sections for business flows crossing several services	you
7	Validate and publish — <code>kb build</code> without <code>--allow-pending</code> , then <code>kb publish --pr</code>	you

3.1 Step 4 is the actual work

The model's draft is a draft, not a fact. It can name the wrong flow, miss a responsibility, or over-claim one. Approving a section you have not read defeats the entire point of the gate.

Read each `svc.*` section's summary next to its code evidence and correct it. Where a responsibility touches a domain standard, cite `doc-id $section` from the hub instead of restating the rule in your own words.

Budget 10–15 minutes per service. That is the honest cost, and it is a one-time cost. After the seed, `-code` regenerates itself and `-svc` accrues a few lines at a time.

For step 6, skip freely. A missing flow beats a guessed one.

3.2 Why the first publish is `--pr` by hand

The seed happens *before* anything has merged to the default branch, so there is no push for `kb-code.yml` to react to.

Use `kb publish --pr`. Plain `kb publish` would tag the commit and poll the intake for up to ten minutes waiting on a tag-triggered CI run this repository does not have, then fail with a misleading error.

That hub pull request is reviewed by an analyst or architect, like any other hub content. `-svc` is curated knowledge; it publishes through review, never auto-merge.

4. Implementing a ticket

Five commands, scaffolded for Claude Code, GitHub Copilot and Cursor.

```
/dev-implement-ticket <ticket>      intake · resolve · ground · placeholders
├── dev-design      — GATE 1: you approve the design
├── dev-plan        — GATE 2: you approve the plan
├── dev-execute     (repeatable, resumable)
└── dev-handover    — GATE 3: you open the PR
                    GATE 4: you merge it
```

`/dev-implement-ticket` is also the orchestrator: given a fresh ticket it runs the first four steps, then invokes the phases in order, detecting how far a ticket already got so it never redoes finished work.

4.1 Pick your entry point

Situation	Run
New ticket, nothing started	<code>/dev-implement-ticket <ticket></code>
Small ticket, the change is obvious	<code>/dev-implement-ticket <ticket></code> — the flow collapses itself; the design is still written, just a bounded one
Design approved, no plan yet	<code>/dev-plan <id></code>
Plan approved, or execution in progress	<code>/dev-execute <id></code>
Code hand-implemented, needs a PR write-up	<code>/dev-handover <id></code>
Lost track of where a ticket stands	<code>/dev-implement-ticket <id></code>
Just checking a citation	<code>kb resolve <file></code>

Every entry point re-checks citation freshness first. The hub may have published since your last session, so a reference that was `ok` yesterday can be `stale` today. Checking only at handover would be too late.

4.2 Where work lives

File	Written by	Purpose
<code>docs/impl/<id>-design.md</code>	<code>dev-design</code>	the design
<code>docs/impl/<id>-plan.md</code>	<code>dev-plan</code>	one task per acceptance criterion, <code>- []</code> checkboxes
<code>docs/impl/<id>-context.md</code>	<code>dev-implement-ticket</code>	resolved-context cache; gitignored, regenerated on demand

State is derived from the first two files, the current branch, and whether a PR is open — **never from conversation history**. Any phase resumes cold in a brand new session.

`kb init` never touches your `docs/impl/` content. It only adds `.gitkeep` and a `.gitignore` so the context cache never lands in a pull request.

4.3 The four gates

Nothing in this pipeline merges or ships without a human.

1. **Design approved** — you sign off before any plan is written.
2. **Plan approved** — you sign off before any code is written.
3. **PR opened** — the agent writes the body; you open it.
4. **Merged** — you review and merge.

The agent does neither of the last two itself.

5. What is enforced, and what is only asked

Be clear-eyed about the difference.

5.1 Machine-enforced

Rule	Enforced by
The PR carries its evidence	<code>kb pr lint</code> in <code>kb-pr-lint.yml</code> on every pull request
Unreviewed knowledge cannot reach the hub	<code>kb build</code> exits 1 while any <code>-svc</code> section is <code>pending</code>
The context cache is CLI-owned	<code>kb resolve --status-only --cache</code> refuses a cache whose version, ref set or resolved block differs from the ticket's

`kb pr lint` fails when a required section is missing, still holds the template's comment, claims verification with no pasted output, has no Verification fence containing the plan's test command, or when `## TDD exemptions` names a class outside `config`, `ci`, `docs`, `style`.

5.2 Prompt-only — stated in every wrapper, measured by nothing

These are rules the skills repeat. Nothing in `kb` enforces or measures them. The only check is the human at the gate — which means you.

- **TDD.** No production code without a failing test observed first, at every step of `dev-execute`. Exempt classes are named in `docs/tdd-exemptions.md` and declared in the plan.
- **Shown verification.** No completion claim without the real command output. `kb pr lint` checks the PR body, not the session.
- **Pinned values, verbatim.** Every standard-derived value comes from the resolved section at its pinned hub version, with a citation comment.
- **The ticket is read-only.** Findings go back to the analyst; the agent never edits the ticket.
- **Gates 1 and 2.** Design and plan approved before the next phase. The `status:` header records it; a human grants it.
- **OPEN (BA)** . An ambiguous acceptance criterion is escalated, never reinterpreted.

6. Working with citations

```
kb resolve tickets/ABC-123.md      # ok | stale | broken, per ref
kb diff hr-handbook --against <rev>
kb get hr-handbook 4.12 --level 13
```

Status	Meaning	What to do
ok	The pinned content still matches what is published	Implement against it
stale	The source was amended after the ticket was written	Run <code>kb diff</code> , then ask the analyst whether the acceptance criteria still hold. Do not silently implement the new version.
broken	The reference no longer resolves	Escalate. The ticket is not implementable as written.

When a value in your code comes from a standard — a format, an enum, a threshold, a field length — take it from the resolved section at the pinned version and leave a citation comment next to it. Six months later that comment is the only thing connecting the constant to its justification.

7. kb code-ingest — what it extracts

```
kb code-ingest          # defaults are usually right
kb code-ingest --db data/app.sqlite # name a database explicitly
kb code-ingest --json    # machine-readable report
```

Seven extractors, organised by **artefact kind rather than programming language**, because Strata is adopted across many project lines.

Extractor	Section prefix	Reads
services	svc.<name>	compose files, Dockerfiles, k8s manifests, solution and workspace files
deps	dep.<ecosystem>	manifests for Python, JavaScript, Java, .NET, Go, PHP, Rust, Swift, Dart
commands	cmd.<purpose>	package scripts, Makefiles, tox, shell scripts, CI run steps
tree	struct.tree	the files Git tracks, so an ignored virtualenv never appears
schema	db.<table>	SQL migrations, Prisma, Alembic, EF, and explicitly named SQLite files
integrations	int.<name>	environment-variable keys only
api	api.<tag>	in-repo OpenAPI and Swagger contracts

`tree` always finds something, so the zero-detection rule is: `kb code-ingest` exits `1` when **no extractor other than `tree`** found anything. A document holding only a folder listing is a misconfiguration, not knowledge.

7.1 Two security properties

These are design properties, not conveniences.

1. **Keys, never values.** The only environment file the integrations extractor opens is an `.env.example`, `.sample` or `.template` — never a real `.env` — and its pattern has no capture group around the value at all, so there is nothing to leak whatever a matched line contains. Compose `environment:` blocks *are* read and do hold real inline values, but only the key survives.
2. **Explicit database input.** SQLite schema is read only from paths you name on the command line. A scratch or fixture database in the repository can never become published company knowledge by accident.

7.2 Stated parser limits

These are limits, not bugs.

- `schema` recognises `CREATE TABLE` and `ALTER TABLE ... ADD COLUMN` in sorted filename order. It does not implement a SQL dialect.
- A `CREATE TABLE` for a name already created in a *different* migration directory is dropped with a warning naming both files, and an `ADD COLUMN` applies only inside the directory of its `CREATE TABLE`. Two independent schemas sharing a table name are never merged into one that exists nowhere.
- `RENAME`, `DROP`, `ADD CONSTRAINT` and `ADD INDEX` warn as out of scope.
- A table name outside the recognised identifier shape is skipped **with a warning naming the file**. The reader counts `CREATE TABLE` keywords and compares against what it recognised, so a skip is never silent.
- EF Core migrations are recognised by table name only.
- `build.gradle` is matched by regex, not parsed as a DSL.
- Maven `<plugins>`, `setup.cfg` `extras_require`, `pnpm-workspace.yaml` and compose `healthcheck:` are not read.

7.3 A destination it did not write is refused

If `.kb/<doc_id>/` already holds Markdown or a manifest and that manifest is missing, unreadable, not titled `... -code knowledge`, or lists a section id outside the seven generated prefixes, `kb code-ingest` exits 1 and writes nothing. There is no override flag — move the document or choose another `--doc-id`.

7.4 Determinism and the dirty-tree caveat

Re-running on the same commit on the same platform produces byte-identical files, which is what makes the CI publish a true no-op on an unchanged tree.

Two honest limits: a few glob matches case-normalise per operating system, so the guarantee is per-platform (CI is always Linux, so what gets published is stable); and the manifest's revision derives from the **HEAD commit** while your working tree may hold uncommitted changes. `kb code-ingest` detects that and prints a `dirty_tree` warning.

8. Keeping knowledge current

Neither document needs new per-ticket discipline from you.

8.1 `--code` needs nothing

`kb-code.yml` re-runs `kb code-ingest` → `kb build` → `kb ci-publish` on every push to the default branch, so it always reflects the current commit.

Trigger	Steps	Publishes
push to default branch, or manual dispatch	code-ingest → build → ci-publish	yes
pull request	build only	never

`--scaffold-svc` is deliberately never passed in CI. CI must never create `pending` content, since that would fail its own build step.

8.2 `-svc` accrues at handover

`dev-handover` runs, for every service a ticket touched:

```
kb svc note payments-api --ticket ABC-123 \
  --title "Add refund endpoint" --refs "payments-spec $4.2"
```

That appends one row to `hist.<service>`. It is deterministic and **idempotent**: re-running for the same ticket and service updates the row rather than duplicating it.

You never hand-edit `hist.*`. It is an append-only log written only by `kb svc note`. Those sections are facts — "this ticket touched this service" — not claims, so they need no review and never block a build.

`kb svc note` exits 1 on any of: no resolvable repo id; a missing `-svc` or paired `-code` document; an unknown service (a typo must not invent one); an unreadable manifest; or a manifest/file desync.

8.3 Drift in responsibility text

Responsibility text can drift, and the system **self-reports rather than self-heals**.

When `kb code-ingest --scaffold-svc` refreshes a service's code evidence and that section's summary is already `reviewed`, it compares the old and new rendered evidence as a whole, whitespace-trimmed string — not a semantic diff over specific fields. **Any** change beyond leading and trailing whitespace flags `stale-risk: svc.<name>` in the report.

A real change trips it — a file or table added or removed. So does a change in nothing but *how* the evidence is rendered, with no code changed underneath. It is not an error, and it never touches the summary. It is a flag that a human should look.

9. Troubleshooting

Symptom	Cause	Fix
<code>kb code-ingest</code> exits 1: nothing detected	Only <code>tree</code> found anything	Check that your compose file, manifests or CI workflows are where the extractors look; name a <code>--db</code> if schema is the point
<code>kb code-ingest</code> exits 1: destination refused	A hand-curated document sits at that doc id	Move it, or use <code>--doc-id</code>
<code>kb build</code> fails in CI on a <code>-svc</code> section	A <code>pending</code> section was committed	Finish the seed: summarize, review, <code>kb approve</code>
<code>kb svc note</code> : "unknown service"	No matching <code>svc.<name></code> in <code>-code</code>	Check the spelling against <code>kb get <repo>-code</code> output; a typo must not invent a service
<code>kb publish</code> hangs, then fails citing the Actions run	Plain <code>kb publish</code> on a repo whose CI has no tag trigger	Use <code>kb publish --pr</code>
<code>kb ci-publish</code> refused	This repo is not in the hub's <code>federation/registry.yaml</code>	Ask the hub maintainer to add it
<code>dirty_tree</code> warning	Uncommitted changes while the manifest revision comes from HEAD	Harmless locally; CI always runs clean
<code>pr-lint</code> fails on a Dependabot PR	The gate never self-skips once required	Expected. Decide whether to keep it required for bot PRs.
A citation resolves <code>stale</code> mid-implementation	The hub published after the ticket was written	<code>kb diff</code> , then ask the analyst. Do not reinterpret the AC yourself.
MCP server unreachable	<code>STRATA_KB_HUB_URL</code> or <code>STRATA_KB_HTTP_TOKEN</code> unset or wrong	Re-run <code>kb mcp-setup</code> (<code>/kb-mcp-setup</code>) — it diagnoses which one, and a bare re-run re-verifies without retyping the token
<code>kb mcp-setup</code> fails with "token was rejected" after the hub maintainer gave you a fresh one	A bare re-run reads the old token straight back out of <code>.env</code> — the prompt only appears when nothing is on disk yet	<code>STRATA_KB_HTTP_TOKEN=<new-token></code> <code>kb mcp-setup</code> , or delete the <code>STRATA_KB_HTTP_TOKEN</code> line from <code>.env</code> and re-run

10. Command summary

Command	Purpose	Exit
<code>kb mcp-setup [--hub-url URL] [--no-verify]</code>	Write the hub's HTTP MCP credentials into <code>.env</code> and verify them	<code>0</code> ok, <code>1</code> no value/probe failed
<code>kb code-ingest [--db p] [--scaffold-svc] [--json]</code>	Extract code structure into <code>-code</code>	<code>0</code> ok, <code>1</code> nothing detected or destination refused
<code>kb svc note <svc> --ticket <id> --title "..."</code>	Append a row to <code>hist.<svc></code>	<code>0</code> ok, <code>1</code> unknown service or missing document
<code>kb build [--strict]</code>	Validate the store	<code>0</code> ok, <code>1</code> error
<code>kb approve <doc> [--section <id>]</code>	Flip corrected sections to <code>reviewed</code>	<code>0</code> ok
<code>kb publish --pr</code>	Open a hub PR (the seed's first publish)	<code>0</code> ok
<code>kb resolve <file></code>	Check a ticket's citations	<code>0</code> ok, <code>1</code> broken, <code>2</code> stale
<code>kb diff <doc> --against <rev></code>	What an amendment changed	<code>0</code>
<code>kb query <text> / kb get <doc> <section></code>	Search and fetch	<code>0</code>
<code>kb pr lint <file></code>	PR evidence gate	<code>0</code> PASS, <code>1</code> FAIL

Slash commands: `/dev-implement-ticket`, `/dev-design`, `/dev-plan`, `/dev-execute`, `/dev-handover`, `/dev-code-seed`.

11. Reserved names

`<repo_id>-code` and `<repo_id>-svc` are reserved for this repository's own code knowledge. A domain document must never take either suffix.

Section prefixes are a published contract that analyst tooling and future tooling key on. Each has exactly one owner:

Prefix	Document	Owner
<code>stact-<name></code> , <code>svc-<name></code> , <code>db-<table></code> , <code>dep-<code></code> , <code>int-<name></code> , <code>api-<step></code> , <code>md-<purpose></code>	<code>-code</code>	extractor
<code>svc-<name></code>	<code>-svc</code>	you (model-drafted)
<code>flow-<name></code>	<code>-flow</code>	you
<code>hist-<name></code>	<code>-hist</code>	kb: svc: note only

`svc.<name>` collides across the two documents **on purpose**. It is the join key that lets a C4 `Container(alias, label, technology, description)` be filled from `-code` (the first three) and `-svc` (the fourth).